Theorems of the Alternative: Duality at Work

Exactly one of the following two statements must hold: For some given matrix A,

(A) $\exists\, x \neq 0$, s.t. $Ax = 0$, $x \geq 0$

(B) $\exists\, p$ s.t. $p^T A > 0$

$$\boxed{A \Rightarrow \sim B}$$

Let $\hat{x}$ be s.t. $\hat{x} \geq 0$, $\hat{x} \neq 0$, $A\hat{x} = 0$.

Since $\hat{x} \geq 0$, $\hat{x} \neq 0$, then if $u > 0$, $u^T \hat{x} > 0$

$\forall p$, $\underbrace{p^T A \hat{x}} = p^T (A\hat{x}) = p^T 0 = 0$ hence $p^T A$ cannot be c.w. strictly +ve $\forall p$, hence $\boxed{\sim B}$

Using Duality

$\sim A$: $Ax = 0,\ x \geq 0 \Rightarrow x = 0$

max: $\mathbb{1}^T x = \sum x_i$

st: $Ax = 0$

$x \geq 0$

Feasible with opt. value $= 0$.

min: $p^T 0$

st: $p^T A \geq \mathbb{1}$

dual feasible $\Rightarrow \exists\ p$ with $p^T A \geq \mathbb{1}$

i.e., $p^T A > 0$.

$\underline{B}$.

Using a separation argument — 1/2

If C is a convex set

and $w \notin C$

Then $\exists$ hyperplane separating w

from C

Not nec true for non-convex

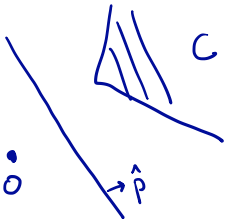
Ex:

Using a separation argument — 2/2

$$C = \{ z : z = Ax, \ \sum x_i \geq 1, \ x \geq 0 \}$$

(A) $\exists \hat{x}$ s.t. $\hat{x} \geq 0$, $A\hat{x} = 0$, $\hat{x} \neq 0$

Equivalent to: $0 \in C$.

(B) $\exists \hat{p}$ s.t. $\hat{p}^T z > 0 \ \forall z \in C$,

$\hat{p}^T 0 = 0$.